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BLOG

# Renewable Energy Sources for HVAC

As we venture into a greener and more sustainable society, we begin to find new ways of powering our homes, businesses, devices, and more. An essential part of the conversation is renewable energy, which helps produce less greenhouse gases and pollution in our environment.

HVAC is an absolute necessity for every building. A steady supply of energy is crucial to heat, cool, and ventilate buildings effectively. Although the majority of electricity and energy in the United States comes from fossil fuels, there is also massive growth in alternative renewable energy technologies. For the HVAC industry, in particular, solar and geothermal energy are best for heating and cooling in residential areas.

## What is renewable energy?

In simple terms, renewable energy is energy production from non-finite resources. The Earth’s supply of oil, fossil fuels, and coal eventually runs out; however, sunlight and wind will not.

## Renewable Energy Types for HVAC Systems

### Solar

As its name suggests, solar energy comes from sunlight. Photovoltaic cells capture the sun’s rays and convert them into electricity through solar panels. Then, the energy is used directly or through the grid to supply power to a building’s HVAC system.

- **Pro:** Most of the Earth’s population has access to sunlight, making solar energy a widely accessible renewable source for HVAC.
- **Con:** Solar panels can save more money in the long run; however, they are expensive to purchase and install.

### Geothermal

Geothermal energy is heat that comes from the rocks and fluids underneath the Earth’s crust. By digging tunnels and creating wells into the ground, steam and hot water can be used to drive turbines connected to electricity generators.

**A geothermal system** consists of an indoor handling unit and pipes that buried in the ground. These pipes (earth loops) and connect from a pump to reinjection well to provide steam. Since the temperature in the Earth’s crust is constant, it provides free energy.

In winter, the fluid that circulates through the system’s earth loop is carried indoors. The indoor unit then transforms this heat to a higher temperature and releases it into the building, acting as a heater. In summer, the geothermal system absorbs heat from the building and transfers it through the earth loop/pump to reinjection well. It’s important to note that geothermal systems do not need periodic maintenance like most HVAC systems. Once installed, the earth loop can last for generations. However, filter changes and coil cleaning can help improve efficiency and longevity as well.

- **Pro:** Geothermal energy is essentially self-replenishing, and has little impact on the surface ecosystems.
- **Con:** Geothermal plants are expensive to build, and certain areas of the world that are prone to earthquakes would not be suitable for geothermal plants.

## Conclusion

As HVAC technologies advance, we begin to find new ways to reduce the negative effects that they create. Renewable energy sources like solar and geothermal energy can produce constant and reliable energy to power our homes, buildings, and businesses. As we begin to use renewable technologies, we will see improvements to our environment and increased efficiency in HVAC systems.

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